

Semantic LaTeX Template

Demo

A book of university notes

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These notes were typeset using $\text{\LaTeX}$ software.

Preface

This short preface demonstrates how the template places optional front-matter sections before the main table of contents.

It is a good place for context, scope, acknowledgements, or reading advice.

To everyone who keeps refining their notes until they become a real book.

Everything should be made as simple as possible, but not simpler.

— ALBERT EINSTEIN

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Part I

Fundamentals

Chapter 1

AVL trees

Binary search tree property (1.1)

An AVL tree satisfies the binary search tree property.

1.1 Balance and height

Definition of Balance factor (1.2)

For each internal node x , the balance factor is

$$\text{bal}(x) = h(\text{left}(x)) - h(\text{right}(x)).$$

Extensionality (1.3)

Two sets are equal if and only if they have the same elements.

Theorem (1.4)

An AVL tree with n nodes has height $O(\log n)$.

Proof:

The key idea is that the minimum number of nodes in an AVL tree of height h satisfies a Fibonacci-style recurrence, so the height grows logarithmically in the number of nodes.

□

Part II

Tools and applications

Chapter 2

Algorithms and diagrams

2.1 Search

Algorithm 1: Search in an AVL tree

Input: Root node r , key k

Output: The node containing k , if it exists

```
1 while  $r \neq nil$  do
2   if  $k = r.key$  then
3     return  $r$ 
4   else if  $k < r.key$  then
5      $r \leftarrow r.left$ 
6   else
7      $r \leftarrow r.right$ 
8 return  $nil$ 
```

2.2 A diagram

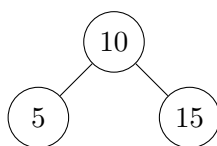


Figure 2.1: Simple AVL diagram drawn directly with TikZ.

Chapter 3

Mathematical notation and references



Lemma

(3.1)

For real x , the expression $|x|$ is non-negative.

3.1 Numbers and delimiters

The commands keep their previous syntax: $\{1, 2, 3\}$, $|-2|$ and $\left\|\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\|$. Unicode maths also supports $\mathbb{R}$, $\mathcal{F}$, $\mathbf{v}$ and $\int_0^1 x^2 dx = \frac{1}{3}$.



Observation

(3.2)

The statement counter continues across sections of the same chapter.

The earlier [definition 1.2](#) and [theorem 1.4](#) belong to [Chapter 1](#). This chapter's [lemma 3.1](#) has its own counter sequence.

Table 3.1: Notation used in the examples.

Notation	Meaning
$ x $	Absolute value
$\ v\ $	Vector norm
$\{1, 2\}$	A finite set

3.2 Code and figures

```
def square(x):  
    return x * x
```

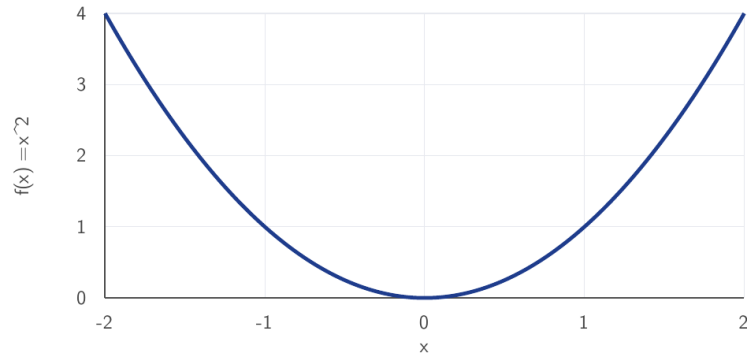


Figure 3.1: The quadratic function on a symmetric interval.

 Tip

(3.3)

PDF and PNG figures compile without enabling shell escape.

3.3 Lists and derivations

- 1 Choose the input.
- 2 Compute the square.
- The badge style matches the original template.
- This list remains a standard LaTeX list.

Derivation:

- 1 $(x + 1)^2 = x^2 + 2x + 1$ *(expansion)*
- 2 $(x + 1)^2 - x^2 = 2x + 1$ *(subtraction)*

Time Complexity Analysis:

Evaluating the square takes constant time.

Appendix A

Supplementary material

Definition of A finite sequence

(A.1)

A finite sequence has a fixed number of terms.

Appendix boxes use letters, as in [definition A.1](#).